

When the Right Signal Fails: Climate Risk, Precautionary Saving, and the Limits of Self-Insurance^{*}

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Abstract

How do households self-insure against climate risk when the underlying distribution is unobserved? I derive climate contrast from a recency-weighted reference-point framework, distinguishing the ordering of recent shocks from their overall variability. In Tanzania, higher contrast is associated with greater assessed shock exposure, more emergency saving, stronger asset buffers, and better food security. The saving relationship extends across 29 low- and middle-income countries. Yet contrast has little predictive content for subsequent drought and climate variability, while conventional variability measures do. Households thus appear to respond to a different climate signal from the one that forecasts these risks.

Keywords: climate risk; income risk; precautionary saving; household saving; informal finance; climate variability; risk perception

JEL codes: D14, D81, O16, Q54

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1 Introduction

How do households form beliefs about economic risk when the underlying probability distribution is unobserved? In many environments, agents face substantial income risk but lack access to formal insurance, reliable forecasts, or long historical records from which to infer stochastic processes (Manski, 2004; Malmendier and Nagel, 2011). Instead, beliefs must be formed from realized sequences of outcomes.

For rainfed agricultural households, climate shocks translate directly into income shocks, providing a setting in which realized sequences of outcomes shape beliefs about risk. Existing work typically measures this risk through variability (Paxson, 1992; Carroll and Samwick, 1997; Alem and Colmer, 2022), yet its empirical relationship with precautionary saving is often weak or signed the “wrong” way (Paxson, 1992; Lugilde, Bande and Riveiro, 2019).

This paper documents a strong mismatch between the climate statistics that predict future risk and those that shape household behavior. Climate variability is genuinely informative about the future: recent variability predicts subsequent drought incidence and climate volatility. Yet households’ assessed climate exposure and precautionary saving instead respond to the *temporal ordering* of recent realizations, which carries little incremental predictive content for future climate outcomes. The statistic that forecasts risk is not the statistic households appear to act on.

To formalize the roles of variability and temporal ordering in shaping households’ climate risk perception, I derive the metrics *climate variability* and *climate contrast* analytically, through a recency-weighted framework in which climate realizations are evaluated relative to a reference point. If realizations are judged relative to the long-run local mean, expected surprise is proportional to *climate variability* only. If instead they are judged relative to the immediately preceding realization (“sequential contrast”), surprise depends not only on dispersion but also on the sequencing of outcomes, as captured by *climate contrast*. These are limiting cases of a single recency-weighted benchmark indexed by a memory parameter, consistent with a growing body of evidence that economic judgments depend on recent reference points and sequential contrasts (Simonsohn, 2006; Bhargava and Fisman, 2014; Hartzmark and Shue, 2018; Jin et al., 2024).

In addition, climate variability may erode households’ ability to act on perceived risk through asset depletion. Precautionary saving requires not only a motive, but also sufficient resources. Under incomplete markets, current asset holdings constrain households’ ability to save (Zeldes, 1989; Deaton, 1991; Carroll and Samwick, 1997); and in climate-exposed agricultural settings, adverse shocks are frequently absorbed through asset sales and other forms of buffer decumulation (Kazianga and Udry, 2006; Carter et al., 2007; Janzen and

Carter, 2019).

To empirically evaluate the role of climate variability and contrast, I combine several household survey datasets with gridded climate data to measure households' exposure to local climate conditions. Climate is measured using the Standardized Precipitation–Evapotranspiration Index (SPEI), which captures variation in water availability relevant for rainfed agriculture (Vicente-Serrano, Beguería and López-Moreno, 2010). Guided by the analytical framework, I construct *climate contrast* as the average absolute year-to-year change in SPEI over recent years, scaled by climate variability over the same window. The measure captures the extent to which climate conditions reverse from one year to the next without scaling mechanically with their overall dispersion. Two locations can experience sequences with the same realized mean and variance while differing sharply in the ordering of shocks.

The main empirical analysis uses two complementary datasets from Tanzania: FinScope district-level repeated cross-sections and the Tanzania National Panel Survey (NPS), which permits within-household comparisons over time. I first examine households' assessment of their climate exposure. A one-standard-deviation increase in climate contrast is associated with a 1.8–2.4 percentage point higher probability of reporting a severe climate shock, conditional on realized climate conditions and the magnitude of each of the previous five years' realizations. Across alternative reference benchmarks, reported shock exposure is most strongly and precisely associated with measures that place greatest weight on the immediately preceding realization, attenuating as the benchmark incorporates more distant history.

Analysis of precautionary behavior shows that a one-standard-deviation increase in climate contrast raises emergency saving by 2.4–2.9 percentage points, while non-emergency saving does not respond. The asset results also help explain why conventional variability measures can be difficult to interpret behaviorally. A one-standard-deviation increase in climate variability is associated with an approximately 20 log point *reduction* in household asset holdings, whereas climate contrast is associated with a 10–12 log point *increase*; both relationships are strongest among initially asset-poor households. Consistent with these buffers having real consumption consequences, climate contrast is also associated with a 1.9–2.1 percentage point reduction in reported food shortages, about 6–7 percent of the baseline rate.

Using harmonized household data from 29 low- and middle-income countries to evaluate external validity, I find that the saving pattern extends beyond Tanzania. In rural areas, a one-standard-deviation increase in climate contrast is associated with a 2.4 percentage point higher probability of saving, about 4 percent of the baseline rate, while conventional climate variability has little predictive power for saving. The association is absent in urban areas,

where livelihoods are less directly exposed to local climate conditions.

A natural alternative explanation is that climate reversals contain information about future climate risk which households respond to. I directly test whether contrast forecasts future climate conditions. It does not: conditional on realized climate exposure, contrast predicts neither subsequent drought nor climate variability, while recent variability predicts both. The statistic households respond to is therefore not the one that best forecasts future risk.

The paper makes three contributions. First, it shows that the temporal ordering of experienced shocks shapes perceived risk and self-insurance, connecting experience-based belief formation (Malmendier and Nagel, 2011; Bordalo, Gennaioli and Shleifer, 2020) and sequential contrast (Simonsohn, 2006; Bhargava and Fisman, 2014; Hartzmark and Shue, 2018; Jin et al., 2024) to economically consequential household behavior. The recency-weighted benchmark nests mean-based and sequential evaluation as limiting cases of a single memory parameter; the reporting response is strongest under short-memory benchmarks and attenuates as the reference horizon lengthens.

Second, it documents a sharp distinction between the statistic households respond to and the statistic that actually forecasts risk: contrast predicts perceived exposure and saving, while variability predicts future drought and volatility. This helps explain why conventional risk measures can be weakly related to precautionary saving (Paxson, 1992; Alem and Colmer, 2022; Abay et al., 2022; Lugilde, Bande and Riveiro, 2019), particularly when realized shocks simultaneously erode households' capacity to save (Zeldes, 1989; Deaton, 1991; Carroll and Samwick, 1997).

Third, it shows that these patterns have economically meaningful consequences for households' short-run food security and their ability to smooth consumption, complementing a large literature on ex-post risk-coping among poor rural households — asset sales, labor supply adjustments, remittances, and child investment (Udry, 1995; Fafchamps, Udry and Czukas, 1998; Kochar, 1999; Suri, 2017; Maccini and Yang, 2009) — by documenting a forward-looking, experience-driven margin of adjustment that appears across diverse institutional and climatic environments in a multitude of low- and middle income countries. Climate projections further show that warming raises objective risk while leaving contrast largely unchanged, potentially widening the gap between risk and the signal households respond to.

The remainder of the paper is organized as follows. Section 2 derives the analytical framework of belief formation and asset-constrained saving, and testable predictions. Section 3 describes the data and empirical strategy. Section 4 tests how climate contrast affects households' memory of climate shocks. Section 5 examines the precautionary saving responses,

the asset-depletion channel, and food security. Section 6 presents cross-country evidence on household saving behavior as a test of external validity. Section 7 tests whether climate contrast forecasts future climate risk, and examines the implications of projected climate change on drought risk and the climate contrast metric. Section 8 concludes.

2 Financial Coping, Climate Risk, and Belief Formation

2.1 Precautionary Saving as the Primary Climate Coping Strategy

Low-income households exposed to climate risk rely on a wide range of coping mechanisms, such as cash savings (Karlan et al., 2017), selling assets (Kazianga and Udry, 2006; Janzen and Carter, 2019), reducing consumption (Nguyen, Nguyen and Grote, 2020), increasing labor supply (Kochar, 1999), and migration (Bohra-Mishra, Oppenheimer and Hsiang, 2014). In environments with limited access to formal insurance and credit, these margins substitute for missing formal risk-sharing mechanisms.

However, descriptive evidence from household panel data indicates that own saving plays a first-order role: among Tanzanian households, nearly *half* report the use of their own savings as their primary coping strategy — substantially more than any other margin (Appendix Table B.1). This motivates focusing on saving as the main precautionary margin of interest.

In addition, rainfed agriculture remains the dominant form of agriculture throughout the developing world (Wani et al., 2009), making rural households in this setting particularly vulnerable to climate shocks (Wasko et al., 2021). Since these households have especially limited access to formal financial markets (World Bank, 2023), and instead rely heavily on informal saving and risk-sharing arrangements such as ROSCAs (Anderson and Baland, 2002; Beaman, Karlan and Thuysbaert, 2014) and mobile money services (Jack and Suri, 2014; Demirguc-Kunt et al., 2018), beliefs about future climate risk may be especially consequential for financial behavior.

2.2 Precautionary Saving under Perceived Income Risk

A central implication of precautionary saving theory is that, holding expected income fixed, greater perceived variance of future income raises desired saving under a *prudent* utility function¹ (Leland, 1968), including standard CRRA and CARA preferences. I therefore

¹ I.e. satisfying $u'''(c) > 0$, a strictly stronger condition than risk aversion ($u''(c) < 0$).

write desired saving today simply as

$$s_0^* = S(\hat{\sigma}_w^2; \theta), \quad \partial S / \partial \hat{\sigma}_w^2 > 0, \quad (1)$$

where $\hat{\sigma}_w^2$ is the household's *perceived* variance of climate-driven agricultural returns and $\theta \in [0, 1]$ is the share of household income exposed to agriculture, mapping that climate risk into overall income risk, with S weakly increasing in θ : households whose income is more exposed to climate, such as rural households, should respond more strongly to a given change in perceived risk. With incomplete credit markets, this precautionary motive is amplified, and saving becomes the primary buffer against adverse realizations (Deaton, 1989).

Importantly, the object governing saving behavior in Equation (1) is *perceived*, rather than objective, climate risk. In practice, households in climate-exposed and low-income settings do not observe climate as a stationary stochastic process with known parameters. Instead, they experience climate as a sequence of realized shocks that directly affect income, without access to long historical records or reliable forecasts (De Longueville et al., 2020; Akponikpè, Johnston and Agbossou, 2010). Beliefs about future climate risk, captured by $\hat{\sigma}_w^2$, must therefore be formed from recent experience rather than inferred from a known probability distribution.

2.3 Experience-Based Belief Formation and Climate Contrast

A growing literature shows that experience-based belief formation generates sensitivity to recent and contrastive realizations even when the underlying distribution is unchanged (Malmendier and Nagel, 2011; Bordalo, Gennaioli and Shleifer, 2020). Consistent with this, Deryugina (2013) shows that beliefs about future climate are shaped primarily by local recent experience rather than by aggregate long-run trends. The implication is that $\hat{\sigma}_w^2$ may depend not only on the distribution of past realizations, but on how much weight the reference point used to evaluate them places on the recent past. A dry year, for example, may be perceived differently after several wet years than after a sequence of similarly dry years.

Consider a farmer whose harvest at time t depends on weather conditions W_t . For tractability, let $\{W_t\}$ be a covariance-stationary process with mean μ_W , variance σ_W^2 , and first-order autocorrelation ρ .

Definition 1 (Recency-weighted reference point and surprise) For $\delta \in [0, 1)$, define the *recency-weighted reference point*

$$B_t(\delta) = (1 - \delta) \sum_{k=1}^{\infty} \delta^{k-1} W_{t-k}, \quad (2)$$

and surprise as $S_t(\delta) = |W_t - B_t(\delta)|$. The parameter δ governs the persistence of memory: larger δ places greater weight on distant realizations. Equivalently, $\delta = e^{-\lambda}$ for decay rate $\lambda \geq 0$, with half-life $\ln 2/\lambda$.

Proposition 1 (Mean-based and sequential surprise as limiting cases) Suppose $\{W_t\}$ is covariance stationary and mean-square ergodic, with Gaussian marginal distribution $N(\mu_W, \sigma_W^2)$. Then $B_t(\delta) \rightarrow \mu_W$ in mean square as $\delta \rightarrow 1^-$, while $B_t(\delta) \rightarrow W_{t-1}$ as $\delta \rightarrow 0$. Consequently,

$$\lim_{\delta \rightarrow 1^-} E[S_t(\delta)] = E[|W_t - \mu_W|] = \sigma_W \sqrt{\frac{2}{\pi}} \equiv \text{Mean-based surprise}, \quad (3)$$

$$\lim_{\delta \rightarrow 0} E[S_t(\delta)] = E[|W_t - W_{t-1}|] \equiv \text{Sequential surprise}. \quad (4)$$

Proof. Appendix A.1. □

The long-memory limit that gives rise to *Mean-based surprise* corresponds to how risk is conventionally defined in the precautionary-saving literature, i.e. measures of dispersion around a long-run mean (Paxson, 1992; Carroll and Samwick, 1997; Alem and Colmer, 2022). At the other extreme, short memory makes the previous realization the relevant comparison point, so surprise is determined by the change from one realization to the next (*Sequential surprise*).

This short-memory case connects directly to a growing body of evidence on *sequential contrast effects*: outcomes are evaluated differently depending on what immediately preceded them, even holding fixed the overall distribution of realizations. Such effects have been documented in commuting choices (Simonsohn, 2006), partner evaluation (Bhargava and Fisman, 2014), financial decision-making (Hartzmark and Shue, 2018), and medical judgments (Jin et al., 2024), alongside broader evidence that economic outcomes are evaluated relative to recent reference points (Benartzi and Thaler, 1995). The recency-weighted framework nests both cases within a single memory parameter. In Section 4, I use intermediate memory half-lives to examine where household assessments empirically lie along this continuum.

Assuming consecutive realizations are jointly Gaussian,² sequential surprise can be decomposed into a component reflecting the overall scale of climate fluctuations and a component reflecting their temporal ordering:

$$\text{Sequential Surprise} = E[|W_t - W_{t-1}|] = \underbrace{\sigma_W \sqrt{\frac{2}{\pi}}}_{\text{Dispersion}} \times \underbrace{\sqrt{2(1-\rho)}}_{\text{Contrast (ordering)}}. \quad (5)$$

² SPEI is constructed by mapping the fitted distribution of the climatic water balance onto a standardized normal scale, so its marginal distribution is approximately Gaussian by construction (Vicente-Serrano, Beguería and López-Moreno, 2010).

Proof. Appendix A.2. □

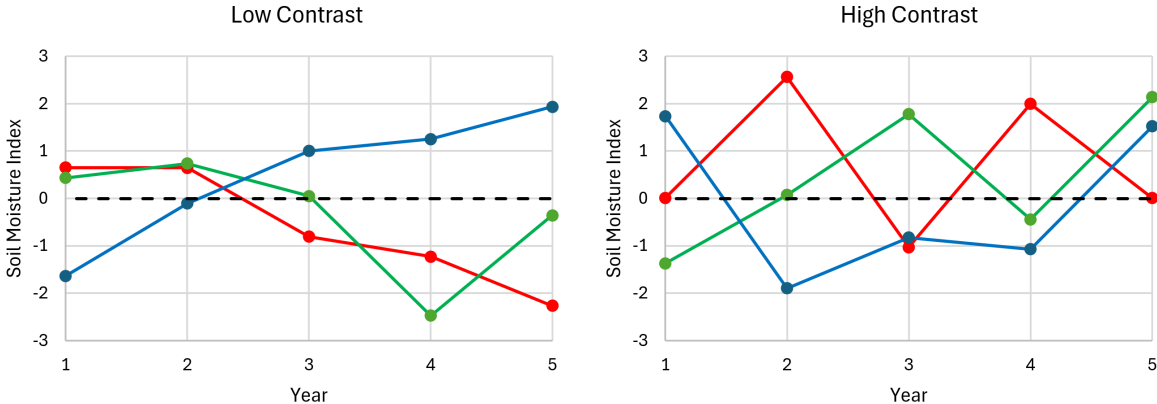
Corollary 1 (Climate Contrast Index) *Normalizing sequential surprise by σ_W yields*

$$\text{Climate Contrast Index} \equiv \frac{E[|W_t - W_{t-1}|]}{\sigma_W} = 2\sqrt{\frac{1 - \rho}{\pi}}, \quad (6)$$

which increases as realizations reverse more sharply from one period to the next and is independent of the scale σ_W .

Equation (5) decomposes sequential surprise into dispersion and temporal ordering. Normalizing by σ_W removes the scale component, yielding a measure of how sharply realizations reverse relative to local variability. This is also consistent with theories of salience, in which contrast is evaluated relative to the scale of the local reference environment (Bordalo, Gennaioli and Shleifer, 2022). Climate contrast therefore isolates the *ordering* of recent experience from its overall *scale*, as illustrated in Figure 1.

Figure 1: Similar Variability, Different Temporal Structure



Notes: Each panel shows three hypothetical five-year climate-index paths with mean zero and comparable overall variance. The left panel shows low-contrast paths with smooth, persistent realizations; the right panel shows high-contrast paths with sharp year-to-year reversals. Standard dispersion-based measures cannot distinguish between these temporal patterns.

Empirically, I use the finite-window analogue of Equation (6): the average absolute year-to-year change in SPEI over the previous five years, divided by the standard deviation of SPEI over the same period. The five-year horizon matches the recall period of past climate shocks in the survey data and is consistent with evidence on relatively rapid decay in responses to past climate shocks (Atreya, Ferreira and Kriesel, 2013; Atreya, Ferreira and Michel-Kerjan, 2015; Bin and Landry, 2013; Moore et al., 2019). Results are robust to exponentially decaying weights over longer histories (Appendix Table E.2), while Section 4 estimates the relationship across the full range of memory half-lives implied by Proposition 1.

2.4 Testable Empirical Predictions

Equation (1) takes perceived risk $\hat{\sigma}_w^2$ as given, while the framework above describes how climate experience is evaluated. I connect the two with the reduced-form mapping

$$\hat{\sigma}_w^2 = g(S_t(\delta)), \quad g' > 0, \quad (7)$$

where δ is the household's effective memory. This requires only that greater experienced surprise raises perceived risk; together with Equation (1), it implies that climate contrast should increase precautionary saving whenever it raises surprise at the relevant memory horizon.

Liquidity constraints. Desired saving need not equal realized saving. Under incomplete markets, households' ability to build a precautionary buffer depends on the assets they already hold (Zeldes, 1989; Deaton, 1991; Carroll and Samwick, 1997). Climate shocks can simultaneously reduce income and force households to sell livestock, draw down savings, or otherwise decumulate assets to smooth consumption (Kazianga and Udry, 2006; Carter et al., 2007; Janzen and Carter, 2019). Realized climate variability may therefore both increase the desired precautionary buffer and erode the resources available to finance it. This provides a complementary explanation for why variability can be weakly or negatively related to observed saving even when the precautionary motive itself is positive (Paxson, 1992; Lugilde, Bande and Riveiro, 2019). The asset results in Section 5.1 provide corroborating evidence for this mechanism.

The framework yields the following empirical predictions:

- *H1 (Assessed exposure):* Conditional on realized climate exposure, greater climate contrast should increase the probability that households report severe recent climate exposure (Section 4).
- *H2 (Precautionary behavior):* Conditional on realized climate exposure, greater climate contrast should increase precautionary saving, but not other forms of saving (Section 5).
- *H3 (Asset buffers):* Greater realized climate variability should be associated with depletion of household asset buffers, particularly among asset-poor households, whereas climate contrast should not exhibit the same depletion pattern (Section 5.1).
- *H4 (Heterogeneity):* The relationships between climate contrast and precautionary saving should be strongest among rural and agriculture-dependent households (Sections 5–6).

3 Data and Empirical Strategy

3.1 Household Panel Data: Tanzania National Panel Survey

To examine both households’ assessed climate exposure and their saving behavior at the household level, I use the four rounds of the Tanzania National Panel Survey (NPS), fielded from 2008/09 through 2014/15 ([World Bank, 2021](#)). The NPS is a longitudinal survey that tracks the same households over time and provides detailed information on consumption, food security, shock exposure, financial behavior, and migration, allowing me to exploit within-household variation in climate exposure over time.

Each survey round asks households whether they experienced a severe shock over the preceding period, including, among other categories, drought and flood. I use this retrospective shock report to test whether the sequencing of recent climate realizations is related to households’ assessed severity of recent climate exposure (Section 4).

The NPS separately elicits households’ saving behavior and, for households that report saving, the purpose of that saving, distinguishing precautionary motives (saving for emergencies) from non-precautionary motives. I use this distinction to test whether climate contrast affects saving specifically along the precautionary margin (Section 5).

The survey also records migration, consumption, and food security, which I use to examine selective relocation and downstream household outcomes.

3.2 Household Finance Data: FinScope Surveys

The main cross-country analysis uses the FinScope National Surveys conducted by the FinMark Trust ([FinMark Trust, 2022](#)). These nationally representative surveys collect detailed information on household financial behavior, including saving, across low- and middle-income countries, and have been fielded since 2006, primarily in Sub-Saharan Africa, South Asia, and Southeast Asia. The primary outcome measures whether the respondent saved in the past 12 months. The surveys further report individual characteristics including age, gender, education, occupation, and urban residence, which are used as controls.

Tanzania district panel. For Tanzania, the cross-country sample includes four waves (2006, 2009, 2013, and 2017); the 2013 wave is available only at the region level and is excluded from the district analysis. I therefore use the three district-georeferenced waves from 2006, 2009, and 2017, yielding 22,103 respondent observations across 169 districts over an 11-year span. These data allow me to exploit within-district variation in climate exposure over time and to examine mechanisms underlying saving behavior: the Tanzania surveys

identify farming households and elicit respondents’ stated motives for saving, which I use to distinguish precautionary saving from saving for other purposes and to study participation in informal savings groups.

Global cross-sectional sample. I combine all available FinScope survey waves that can be georeferenced at the admin-1 level or finer, excluding small island states and surveys without spatial identifiers. The resulting sample consists of 223,670 respondents across 29 countries and 43 survey waves spanning the period 2006–2022. Figure B.1 illustrates the geographic coverage of the sample, and Table B.3 lists included countries and survey years.

3.3 Climate Data

Climate conditions are measured using the Standardized Precipitation–Evapotranspiration Index (SPEI) (Vicente-Serrano, Beguería and López-Moreno, 2010). SPEI combines precipitation and temperature-driven evapotranspiration to capture net water availability and is widely used in drought monitoring and climate impact studies (Beguería et al., 2014). The index is standardized relative to local historical conditions, with mean zero and unit variance, facilitating comparisons across locations and time.

I use accumulated 12-month SPEI values, which summarize cumulative water balance over the most recent agricultural season, and which has been found to be the most relevant for rainfed agricultural production in low-income settings (Kubik and Maurel, 2016). SPEI values are computed over the 12 months preceding the survey interview date and aggregated to the region (admin-1), or district (admin-2) level depending on data availability.

Based on the definitions in Section 2.3, I construct *Climate Variability* as the standard deviation of annual SPEI values, σ_W , and *Climate Contrast* as the average absolute year-to-year change in SPEI, normalized by σ_W over the same window. Because it is normalized by variability, climate contrast does not scale with the amplitude of the climate shocks.

Throughout the analysis, both climate variability and climate contrast are standardized to have mean zero and unit variance within the estimation sample, so that regression coefficients are directly interpretable as the effect of a one-standard-deviation change in each measure.

3.4 Empirical Strategy

The empirical analysis first examines whether the sequencing of recent climate realizations affects households’ assessed climate exposure and precautionary behavior in Tanzania, where panel data allow me to exploit within-household variation over time. I then test whether the saving relationship generalizes across 29 low- and middle-income countries, treating the

cross-country analysis as evidence on external validity rather than as a second identification strategy. Table B.2 reports summary statistics for the FinScope and NPS samples.

The main specifications relate household outcomes to recent climate variability and climate contrast, conditional on contemporaneous and longer-run climate conditions. SPEI is standardized relative to each location’s historical climate distribution, and both climate measures vary over time within locations. In Tanzania, district and household fixed effects identify the coefficients from within-unit changes in climate exposure over time. In the global sample, country fixed effects absorb persistent cross-country differences, while long-run local climate controls separate recent regional exposure from persistent spatial differences in climate; regional cross-sectional variation within countries therefore remains part of the identifying variation there.

A causal interpretation requires that these within-unit changes in recent climate experience be conditionally exogenous. The main concern is selective migration in response to past climate conditions, particularly in the repeated cross-sections, where changes in sample composition could mimic behavioral responses. I test this using regional migration data from Niva et al. (2023) and the Tanzania household panel and find no evidence that climate contrast predicts migration. The interpretation also requires that contemporaneous climate conditions affect agricultural income, which I validate in Section 3.5, and that past climate contrast does not affect saving through channels other than perceived income risk. The latter is an exclusion restriction. Throughout, I condition on recent and long-run climate conditions and use placebo outcomes and samples where the proposed climate-risk mechanism should be absent to probe alternative explanations.

To validate the income channel, I estimate

$$Y_{kcy} = \alpha_c + \lambda_y + \beta \text{SPEI}_{cy}^{12m} + \gamma \text{Climate Variability}_{cy}^{5y} + \delta \text{Climate Contrast}_{cy}^{5y} + \mathbf{X}'_{kcy} \Gamma + \varepsilon_{kcy}, \quad (8)$$

where Y_{kcy} denotes log agricultural production or yield for crop k in country c and year y , weighted by each crop’s share of national production, and $\mathbf{X}$ includes lagged climate realizations. Contemporaneous SPEI strongly predicts agricultural production, while past climate variability and contrast do not once current conditions are held fixed.

The main household-level specification is

$$Y_{icry} = \alpha_c + \lambda_y + \beta \text{SPEI}_{ry}^{12m} + \gamma \text{Climate Variability}_{ry}^{5y} + \delta \text{Climate Contrast}_{ry}^{5y} + \mathbf{X}'_{icry} \Gamma + \varepsilon_{icry}, \quad (9)$$

where Y_{icry} denotes the household outcome for individual i in region r , country c , and year y . In the Tanzania analyses, country fixed effects are replaced by district or household fixed

effects to exploit within-unit changes in climate exposure. Standard errors are clustered at the level of climate assignment (Abadie et al., 2023), accounting for spatially shared and serially correlated climate shocks.

Finally, households might respond to climate contrast because sharp reversals themselves predict future climate risk rather than because contrast shapes perceived risk. Section 7.1 tests this alternative directly.

3.5 Validation: Climate Conditions and Agricultural Income

I validate the income channel using FAOSTAT production data described in Appendix C. Consistent with a large literature on rainfed agriculture, contemporaneous climate conditions strongly predict agricultural production: a one-standard-deviation increase in 12-month SPEI raises output by approximately 7 percent among countries covered by the FinScope sample, while drought ($\text{SPEI} < -1$) reduces it by roughly 14 percent, comparable to household-level estimates in Kubik and Maurel (2016) (Appendix Table C.1). By contrast, lagged climate variability and climate contrast do not predict current production once contemporaneous conditions are held fixed. This supports interpreting these measures as shifters of perceived future income risk rather than direct determinants of current income.

4 Climate Contrast and Perceived Climate Exposure

I first test whether shock ordering affects households’ assessment of recent climate exposure using the NPS panel. The NPS records whether households report having experienced a severe climate shock in the recent past. By observing repeated reports by the same households over time, I exploit changes in the *ordering* of recent climate realizations faced by the same household, with household fixed effects absorbing persistent reporting tendencies and local conditions.

Because reported climate shocks include both droughts and floods, I condition on the absolute value of the Standardized Precipitation–Evapotranspiration Index (SPEI), which captures deviations in water availability in either direction. This absorbs the magnitude of recent climate realizations so that the estimated effect of climate contrast is not driven by the intensity of realized shocks. All specifications additionally control for year-by-year climate conditions over the previous five years, helping distinguish the contrast association from variation in the cumulative intensity of shocks.

Table 1 reports the results. Across specifications, climate contrast has a positive and statistically significant effect on reported severe climate shocks. The estimates are stable across

Table 1: Climate Contrast and Subjective Assessment of Climate Shocks

	Reports experiencing severe climate shock in the past 5 years							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Climate Variability (5y)	0.021 (0.015)		0.036** (0.017)		0.018 (0.022)		0.034 (0.024)	
Climate Contrast (5y)		0.018*** (0.006)	0.023*** (0.007)	0.023*** (0.006)		0.019* (0.010)	0.024** (0.011)	0.024** (0.010)
FE Specification	District	District	District	District	HH	HH	HH	HH
Survey Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Climate Shock Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
20-bin Variability FE	No	No	No	Yes	No	No	No	Yes
Mean DV	0.268	0.268	0.268	0.268	0.285	0.285	0.285	0.285
Observations	22,680	22,680	22,680	22,680	17,052	17,052	17,052	17,052
Clusters	140	140	140	140	140	140	140	140

Note: *Reports climate shock severity* is an indicator equal to one if the household reports experiencing a severe climate shock (drought or flood) during the previous five years, and zero otherwise. *Abs(SPEI 12m)* is the absolute value of the Standardized Precipitation–Evapotranspiration Index over the past 12 months, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include the absolute values of annual SPEI realizations for each of the previous five years (*Climate Shock Controls*), as well as survey year and month fixed effects. Variability-bin FE are fixed effects for 20 quantile bins of five-year climate variability, allowing for a very flexible relationship between climate variability and the outcome. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

both district and household fixed effects specifications and remain robust to conditioning on realized climate conditions and climate variability.

In the baseline specifications, a one-standard-deviation increase in climate contrast raises the probability of reporting a severe climate shock by approximately 1.8 to 2.4 percentage points. Against baseline prevalence of 26.8 percent in the district-fixed-effects sample and 28.5 percent in the household-fixed-effects sample, this corresponds to about 6–9 percent of the mean. The magnitude is economically meaningful and comparable to the effects observed later for precautionary saving and food security.

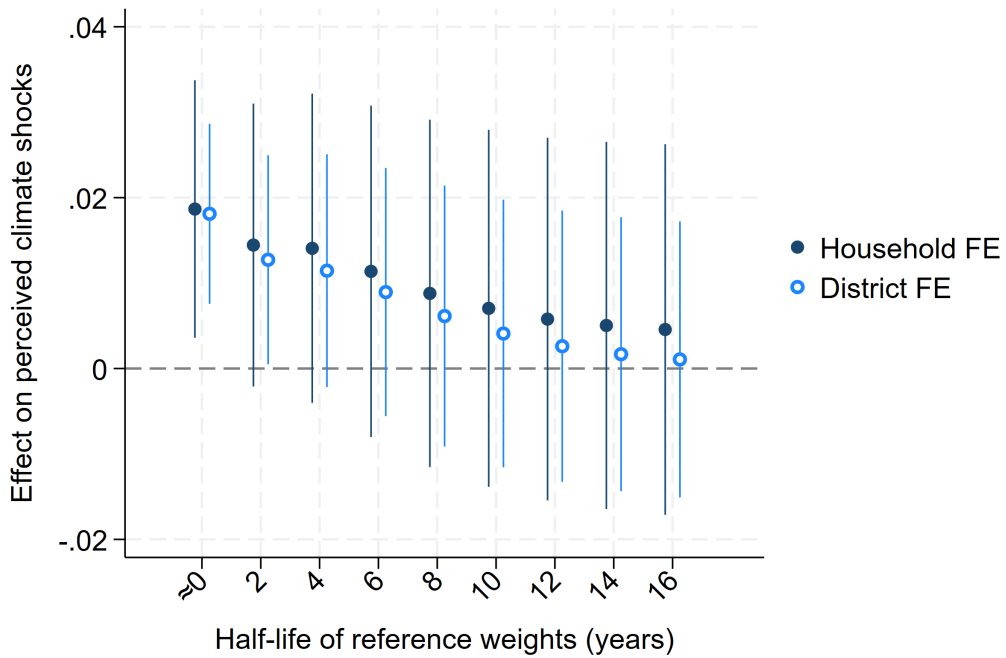
By contrast, climate variability has less robust effects across specifications. While it enters positively in these specifications, the estimates are less stable and sensitive to the inclusion of household fixed effects, indicating that dispersion-based measures of climate risk are weaker predictors of assessed shock severity than the sequencing of climate shocks.

To further examine the mechanism underlying these results, I construct a coefficient profile motivated by Proposition 1. In addition to testing only the two polar cases of mean-based and sequential surprise, I construct a family of recency-weighted climate experience measures indexed by the memory parameter δ of Definition 1, with past realizations entering

with exponentially decaying weights. The half-life of the weighting scheme, $\ln 2/\lambda$ for $\delta = e^{-\lambda}$, determines the effective memory length: short half-lives place most weight on recent realizations and approximate sequential contrast ($\delta \rightarrow 0$), while long half-lives converge toward evaluation relative to long-run climate conditions ($\delta \rightarrow 1^-$).

Figure 2 plots this coefficient across the δ -continuum. The effect is largest and most precisely estimated at short half-lives, consistent with sequential contrast driving the climate contrast effect, and declines steadily as the reference benchmark lengthens toward long-run climate conditions.

Figure 2: Sensitivity of Perceived Climate Shocks to Reference Memory Length



Notes: This figure shows regression estimates of the effect of climate experience on perceived severe climate shocks as the half-life of the exponential weighting scheme used to construct the reference benchmark varies. Short half-lives place greater weight on recent realizations and approximate sequential contrast, while long half-lives approximate evaluation relative to long-run conditions. All measures are normalized by local climate variability to hold constant the scale of climate variation across specifications. Solid markers report point estimates and vertical lines show 95% confidence intervals.

Robustness and Placebo Tests. These findings are robust to a wide range of alternative measures of recent climate exposure, including controls for extreme flood and drought realizations as well as the frequency of climate events over the previous five years (Appendix Table E.1). Results are also robust to accounting for spatial correlation across districts (Appendix Table E.4).

A further prediction of the economic-risk interpretation is that the relationship between climate contrast and perceived risk should be stronger where local climate conditions trans-

late more directly into household economic exposure. Consistent with this, the effect of climate contrast on reported climate shocks is concentrated among rural households and is statistically indistinguishable from zero in urban areas (Appendix Table D.7).

In addition, climate contrast does not predict the reporting of severe non-climate shocks—such as crime, incarceration, or household breakup (Appendix Table E.6).³ These tests weigh against the interpretation that climate contrast merely captures general shock-proneness or reporting behavior.

5 Precautionary Saving and Food Security: Panel Evidence from Tanzania

I next test whether within-household changes in climate contrast affect precautionary saving, conditional on contemporaneous climate conditions.

Table 2 reports the results. Panel A shows that climate contrast is positively associated with precautionary (emergency) saving in all four specifications and statistically significant in the first three. A one-standard-deviation increase in climate contrast raises the probability of saving for emergency purposes by approximately 2.4 to 2.9 percentage points; the household-fixed-effects estimate is 2.5 percentage points but is not statistically significant. The point estimates are similar across the FinScope district-level repeated cross-sections and the NPS household panel, although the within-household estimate is less precise, reflecting the loss of observations when household fixed effects restrict identification to households observed in multiple survey waves.

Climate variability is not statistically distinguishable from zero in these saving specifications. This could be either because it less robustly affects perception of climate risk, or because it depletes assets and thereby limits accumulation of cash saving (evidence for which I present in Section 5.1).

The estimated contrast coefficients for non-precautionary saving motives (Panel B) are close to zero and precisely estimated across specifications. This pattern supports a precautionary interpretation of the emergency-saving response.

Appendix Tables D.4–D.5 provide complementary evidence that climate contrast increases the stated importance of emergency saving and the receipt, but not sending, of mobile-money transfers (Jack and Suri, 2014; Suri, 2017).

Similar to the results for perceived climate risk in Section 4, the saving results are robust

³ The non-climate shock index is constructed as an indicator for whether the household reports at least one severe shock in the following categories: hijacking/robbery/burglary/assault, incarceration, death of a household member, death of another family member, or household breakup. Shock categories plausibly affected by climate conditions—such as food price increases, death of livestock, and crop pests—are excluded.

Table 2: Climate Contrast and Saving Motives among Tanzanian Farmers and Mobile Money Users

	Saving by Motive			
	(1)	(2)	(3)	(4)
Panel A: Precautionary saving				
SPEI (12m)	−0.054** (0.021)	−0.054** (0.021)	−0.067** (0.027)	−0.036 (0.045)
Climate Variability (5y)	0.024 (0.020)	0.027 (0.020)	−0.009 (0.023)	0.041 (0.036)
Climate Contrast (5y)	0.029** (0.012)	0.029** (0.012)	0.024** (0.011)	0.025 (0.024)
Observations	7,205	7,205	7,353	2,714
Clusters	129	129	135	119
Panel B: Non-precautionary saving (Placebo)				
SPEI (12m)	0.018 (0.014)	0.019 (0.014)	−0.005 (0.015)	−0.017 (0.019)
Climate Variability (5y)	−0.033** (0.014)	−0.033** (0.014)	0.007 (0.009)	0.032* (0.016)
Climate Contrast (5y)	0.005 (0.009)	0.005 (0.009)	−0.001 (0.005)	0.002 (0.010)
Observations	8,040	8,040	7,353	2,714
Clusters	129	129	135	119
Sample	FinScope	FinScope	LSMS	LSMS
HH controls	No	Yes	No	No
Type of FE	District FE	District FE	District FE	Household FE
Year FE	Yes	Yes	Yes	Yes

Note: Panel A's dependent variable indicates saving for emergencies; Panel B indicates non-emergency saving motives. Columns (1)–(2) use Tanzania FinScope respondents whose primary occupation is farming. Columns (3)–(4) use NPS respondents who report using mobile money; for these columns, non-emergency saving covers everyday expenses or a large purchase and excludes respondents reporting emergency saving. *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is the average absolute year-to-year change in SPEI over the previous five years, divided by Climate Variability (5y). All regressions include moving averages of SPEI over the previous five years. Household controls in the FinScope regressions include age, education-level fixed effects, an urban indicator, and a female indicator. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

to allowing climate shocks to be spatially correlated across nearby districts, rather than only within a district. Appendix [Table E.4](#) reports Conley spatial-HAC standard errors at bandwidths of 50–400km alongside the baseline district-clustered standard errors. The results are also stable when applying exponentially decaying weights to past realizations instead of uniform weights over a fixed 5-year window, consistent with the literature on memory decay after natural disasters ([Atreya, Ferreira and Michel-Kerjan, 2015](#)) (Appendix [Table E.2](#)).

5.1 Asset Buffers and the Depletion Channel

If realized variability simultaneously raises desired saving and erodes households’ resources, as argued in Section 2.4, its net effect on observed saving is ambiguous. [Table 3](#) examines this argument directly, using self-reported household asset holdings from the Tanzania panel.

Columns (1)–(2) show that climate variability has a large, negative, and precisely estimated effect on log assets: a one-standard-deviation increase in climate variability reduces log assets by approximately 0.20–0.21 log points, holding fixed contemporaneous climate conditions and district or household fixed effects. Climate contrast has the opposite sign, raising log assets by 0.10–0.12 log points. This is the asset-level analogue of the saving results in [Table 2](#): variability is associated with erosion of the resources households would otherwise draw on to self-insure, while contrast is not.

Columns (3)–(4) interact both climate measures with positive asset holdings in the first wave. The district-fixed-effects estimates suggest that variability’s negative effect is concentrated among households starting from zero, for whom it reflects foregone accumulation rather than literal depletion, and that contrast’s positive effect is concentrated among households without an initial buffer; the household-fixed-effects interaction estimates are less precise. This is consistent with liquidity constraints binding most strongly when self-insurance capacity is initially thin.

One interpretation of these results, following the discussion in Section 2.4, is that variability raises the precautionary motive while simultaneously eroding the resources needed to translate that motive into saving, whereas contrast—since it does not scale with the magnitude of the shocks—can raise perceived risk without the same offsetting depletion effect.

5.2 Consumption Smoothing and Buffer-Stock Behavior

Buffer-stock behavior entails both accumulating resources in favorable states and drawing them down when adverse conditions materialize. Because the NPS survey waves largely occurred during dry periods (mean SPEI close to -1), observed consumption behavior mostly reflects households drawing down previously accumulated buffers rather than building them

Table 3: Climate Variability and Household Asset Buffers

	Log assets			
	(1)	(2)	(3)	(4)
Climate Variability (5y)	−0.206*** (0.053)	−0.202*** (0.056)	−0.220*** (0.054)	−0.206*** (0.057)
Variability × Positive assets			0.125* (0.064)	0.226 (0.195)
Climate Contrast (5y)	0.095** (0.037)	0.105** (0.048)	0.117*** (0.040)	0.107** (0.049)
Contrast × Positive assets			−0.112** (0.044)	−0.115 (0.088)
Type of FE	District FE	Household FE	District FE	Household FE
Survey Year FE	Yes	Yes	Yes	Yes
Observations	26,819	21,341	26,815	21,337
Clusters	140	140	140	140

Note: *Log assets* is $\ln(1 + \text{assets})$, where *assets* is the self-reported value of household asset holdings. *Positive assets* is an indicator equal to one if the household reports positive asset holdings in the first survey wave. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) *Climate Variability (5y)*. All regressions include moving averages of SPEI over the previous five years and survey year fixed effects. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

up. Consistent with this, climate contrast is associated with higher non-food expenditure in the baseline specifications (15–25 log points per standard deviation), and this association is concentrated in dry periods and reverses in wet periods — the interaction of climate contrast with contemporaneous SPEI is negative and precisely estimated (Appendix Table D.6). Climate variability shows the opposite state-dependence, positive in wet periods and negative in dry periods, consistent with households exposed to variability lacking sufficient buffers and adjusting consumption more directly in response to current income. This state dependence is consistent with active buffer-stock management.

5.3 Food Security

I next test whether the precautionary response to climate contrast translates into improved food security. Table 4 measures food security as the absence of reported food shortages. While not a comprehensive welfare measure, food shortages capture an economically important margin in a setting where many rural households face recurrent food insecurity.

Climate contrast is associated with fewer food shortages. A one-standard-deviation increase reduces the probability of a reported shortage by 1.9–2.1 percentage points, about 6–7 percent of the 29–30 percent baseline rate. The district-fixed-effects estimate is statistically significant; the similarly sized household-fixed-effects estimate is less precise. Climate

variability shows the opposite pattern, predicting more food shortages.

Unlike the consumption response, the relationship between climate contrast and food security does not depend strongly on contemporaneous climate conditions. This suggests that buffers accumulated in earlier periods help protect essential consumption across adverse states, rather than only during particular contemporaneous climate realizations.

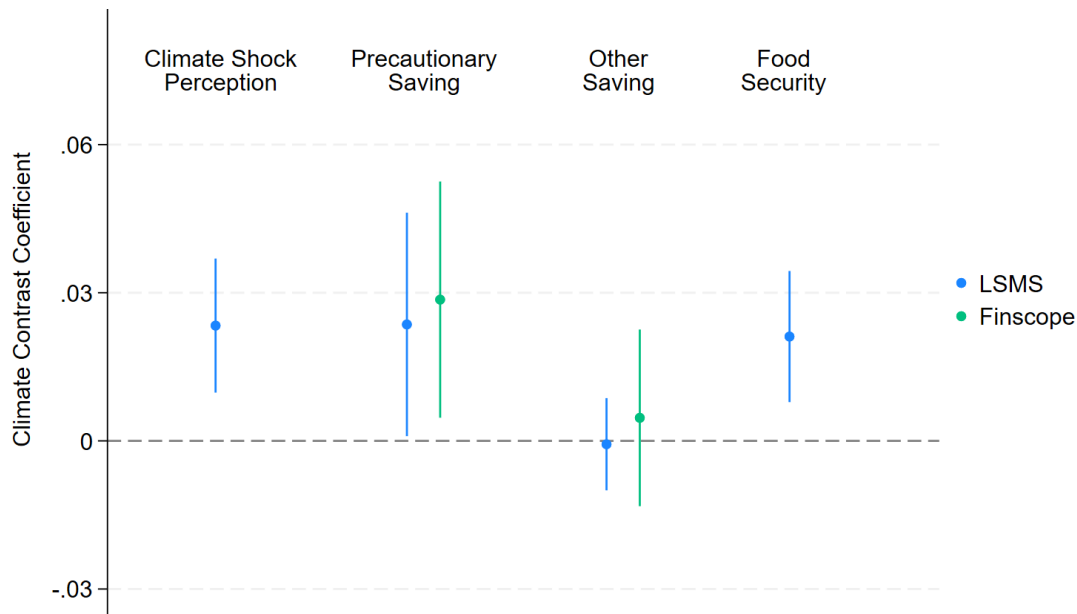
Table 4: Climate Contrast and Food Security

	Food shortage	
	(1)	(2)
SPEI (12m)	0.026* (0.014)	0.026 (0.020)
Climate Variability (5y)	0.048*** (0.016)	0.038 (0.024)
Climate Contrast (5y)	-0.021*** (0.007)	-0.019 (0.012)
Type of FE	District FE	Household FE
Survey Year FE	Yes	Yes
Mean DV	0.301	0.290
Observations	21,255	15,531
Clusters	140	140

Note: *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five years. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Taken together, the Tanzania results reveal a coherent pattern: greater climate contrast raises assessed shock exposure and precautionary saving and is associated with fewer food shortages, without increasing non-precautionary saving. Climate variability instead predicts weaker asset accumulation and more food shortages. The contrast coefficients are strikingly similar in magnitude across assessed exposure, precautionary saving, and food security (Figure 3). Although these estimates do not identify mediation, their alignment is consistent with households building precautionary buffers in response to contrast and using them to protect consumption.

Figure 3: The Effect of Climate Contrast on Assessed Shock Exposure, Saving, and Food Security



Notes: Points report coefficients and 95% confidence intervals from regressions of each outcome on climate contrast, controlling for realized climate conditions and fixed effects.

6 Global Evidence on Climate Contrast and Saving

I next test whether the Tanzania saving result generalizes across 29 low- and middle-income countries. The FinScope surveys are predominantly cross-sectional, so I treat this as evidence on external validity rather than within-household identification. Identification comes from short-run deviations in contrast relative to longer-run climate conditions within countries, hence omitted-variable bias would require these short-term climate-driven deviations to covary systematically with unobserved determinants of saving.

Table 5 reports the cross-country estimates alongside the Tanzania motive-specific results. In the rural sample, a one-standard-deviation increase in climate contrast is associated with a 2.4 percentage point increase in the probability of saving, closely matching the Tanzania precautionary-saving estimates, while climate variability is small and statistically insignificant. The contrast coefficient is stable to controls for long-run climate conditions and household characteristics and is close to zero in urban areas. This rural–urban heterogeneity is consistent with the mechanism operating through exposure to climate-dependent income.

6.1 Timing and State Contingency: Saving in Wet and Dry Years

A central implication of precautionary saving models is that households accumulate buffers during favorable states and decumulate them when income falls. If climate contrast operates

Table 5: Climate Contrast and Saving in Tanzania and the Cross-Country Sample

	Emergency saving			Other saving	Any saving			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Climate Variability (5y)	0.021		0.027	– 0.033**	–0.014		–0.010	–0.004
	(0.021)		(0.020)	(0.014)	(0.009)		(0.009)	(0.008)
Climate Contrast (5y)		0.027**	0.029**	0.005		0.025***	0.024***	0.002
		(0.012)	(0.012)	(0.009)		(0.007)	(0.007)	(0.006)
Sample	Farmers	Farmers	Farmers	Farmers	Rural	Rural	Rural	Urban
Location FE	District	District	District	District	Country	Country	Country	Country
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HH controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean DV			0.296	0.276			0.605	0.628
Observations	7,205	7,205	7,205	8,040	144,191	144,191	144,191	76,758
Clusters	129	129	129	129	368	368	368	358

Note: Columns (1)–(4) reproduce the Tanzania FinScope motive-specific estimates for farmers; columns (5)–(8) use FinScope repeated cross-sections from 29 countries and measure any saving in the past 12 months. *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the district (admin 2) level in columns (1)–(4) and region (admin 1) level in columns (5)–(8). *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is the average absolute year-to-year change in SPEI over the previous five years, divided by Climate Variability (5y). The cross-country specifications include five- and thirty-year climate controls and household controls for age category, education level, urban residence, and gender. Standard errors are clustered at the corresponding climate-assignment level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Climate Contrast and Saving in Wet and Dry Years

	Saved in past 12 months		
	All years (1)	Wet year (2)	Dry year (3)
SPEI (12m)	0.005 (0.012)	0.040** (0.018)	0.003 (0.024)
Climate Variability (5y)	-0.010 (0.009)	-0.009 (0.015)	-0.016 (0.016)
Climate Contrast (5y)	0.024*** (0.007)	0.038*** (0.013)	0.004 (0.010)
Sample	Rural	Rural	Rural
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
HH controls	Yes	Yes	Yes
Mean DV	0.605	0.622	0.590
Observations	144,191	66,050	78,141
Clusters	368	164	237

Note: *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the region (admin 1) level. *Wet year* refers to observations with $SPEI(12m) > 0$, while *Dry year* refers to observations with $SPEI(12m) \leq 0$. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five and thirty years, as well as Climate Variability and climate contrast measured over the previous thirty years. Household controls include age category, education level, an urban indicator, and a female indicator. Standard errors are clustered at the region (admin 1) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

on anticipated risk rather than on contemporaneous income, saving should increase primarily in periods when current conditions are favorable.

Table 6 supports this prediction. The positive effect of climate contrast on saving is substantially larger in wet years than in the pooled specification, while in dry years the effect is small and statistically indistinguishable from zero.

This pattern is consistent with households building precautionary buffers when current conditions are favorable rather than during periods of stress. Climate Variability’s point estimate is negative in dry years, though imprecisely estimated, offering only suggestive evidence that households struggle to accumulate buffers when both long-run uncertainty and current conditions are unfavorable.

6.2 Saving Instruments and Informal Finance

I next examine through *which* financial instruments the precautionary response to climate contrast operates. Table D.8 decomposes saving behavior by instrument type. The response operates predominantly through informal channels.

In wet years, climate contrast is positively associated with saving through informal groups and family or friends, while the formal-saving effect is small. In dry years, formal saving is positive and significant, informal saving is negative at the 10 percent level, and family-or-friend saving is near zero. These estimates indicate a state-dependent change in which instruments households use. The results are consistent with informal arrangements playing an important role in favorable conditions and with households substituting across instruments when conditions are dry. One limitation is that this only measures saving at the extensive margin, and cannot be used to estimate intensive effects on account balances.

6.3 Heterogeneity and Robustness

Heterogeneity by education and gender. The response is concentrated among less-educated households (Table D.2), consistent with greater exposure to climate-dependent income. I find no systematic gender differences (Table D.3).

Robustness and alternative explanations. Climate contrast does not predict migration in either the regional migration panel (Niva et al., 2023) or the Tanzania household panel (Table D.1); the estimates are small and statistically insignificant.

The global results are robust to conditioning on alternative climate controls, including measures of extreme drought and flood exposure (Table E.3). The results are also not driven by any single country. Iteratively excluding one country at a time yields stable and statistically significant coefficients in a narrow range (0.023–0.031), indicating that the findings reflect a broad pattern rather than a small number of influential observations.

Since identification in the largely cross-sectional global sample is weaker, I follow Oster (2019) and assess sensitivity to selection on unobservables by comparing estimates with and without household controls. The inclusion of controls raises the R^2 from 0.12 to 0.14 while reducing the climate contrast coefficient only modestly (from 0.032 to 0.027). Under standard assumptions, the implied degree of selection on unobservables required to eliminate the effect is large ($\delta \approx 9$), indicating that the results are unlikely to be driven by omitted variable bias.

Table E.5 tests whether low weather-station coverage attenuates the global estimates through classical measurement error (Auffhammer et al., 2013). Splitting the sample by above- or below-median rainfall- and temperature-station density shows that the climate contrast estimate is larger and significant above the median but small and insignificant below it, while for temperature-station density it remains positive and significant on both sides. This is consistent with precipitation being more spatially heterogeneous and therefore requiring denser gauge networks for accurate interpolation, whereas temperature can be interpolated more reliably from sparser networks. Under classical measurement error,

attenuation biases coefficients toward zero, so if anything the reported estimates are a lower bound on the true effect.

Together with the panel evidence from Tanzania, these findings indicate that the association between recent climate ordering and precautionary saving is not specific to a single setting, and that it appears where the underlying exposure mechanism should be strongest.

7 Objective Risk, Information, and Climate Change

7.1 Does Climate Contrast Predict Future Climate Risk?

The most natural alternative interpretation of the results is that year-to-year climate reversals convey valuable information about future climate risk — signaling changes in persistence, volatility, or drought regimes. Under this interpretation, households responding to climate contrast would be updating beliefs rationally about an objectively riskier future, rather than responding to the contrastive structure of recent experience itself, as the belief-formation channel of Section 2.3 implies.

To test this, I move to the underlying gridded SPEI data and regress four forward-looking climate outcomes on climate contrast, measured at the 0.5° grid-cell level, controlling for current 12-month SPEI, five-year climate variability, grid-cell fixed effects, and year fixed effects. The outcomes span both the level and variance of future conditions over both the immediate and short term: absolute climate deviations and drought incidence over the next year, and realized climate variability and any drought occurrence over the next five years.

Table 7 reports the results. Across specifications, climate contrast has little incremental predictive content for the future climate outcomes examined here. The estimated coefficients are small and precisely estimated around zero, at both short and medium horizons, and for both level and variance outcomes.

By contrast, current climate variability strongly predicts future climate outcomes at both the immediate and short-term horizons, consistent with standard persistence in the underlying climate process. This asymmetry is the mirror image of the behavioral asymmetry documented in Sections 4–6: while variability carries forward-looking information about the climate process, it does not appear to drive household behavior. Conversely, contrast drives household behavior, but has little predictive content for the future climate outcomes examined here. This favors an experience-based interpretation over the hypothesis that contrast is simply an informative signal of future climate risk.

Table 7: Climate Variability, Climate Contrast, and Future Climate Risk

	Climate risk over next year		Climate risk over next 5 years	
	Abs(SPEI)	Drought	Climate Variability	Any drought
	(1)	(2)	(3)	(4)
SPEI (12-month)	−0.005 (0.008)	−0.008 (0.005)	−0.016*** (0.004)	−0.034*** (0.006)
Climate Variability (5y)	0.061*** (0.010)	0.020*** (0.005)	0.015** (0.007)	0.018** (0.009)
Climate Contrast (5y)	−0.001 (0.001)	−0.000 (0.001)	−0.003 (0.003)	−0.004 (0.002)
Cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Mean DV	0.763	0.193	0.722	0.561
Observations	183,389	183,389	161,733	161,733
Clusters	391	391	391	391

Note: *Abs(SPEI)* is the absolute value of the Standardized Precipitation–Evapotranspiration Index over the past 12 months. *Drought* is an indicator equal to one if SPEI is below −1, and zero otherwise. *SPEI (12m)* is the signed Standardized Precipitation–Evapotranspiration Index over the past 12 months. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All independent variables are lagged one year relative to the outcome. All variables are measured at the 0.5° grid-cell level. Standard errors are clustered at the Admin 1 level corresponding to the FinScope survey data aggregation. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

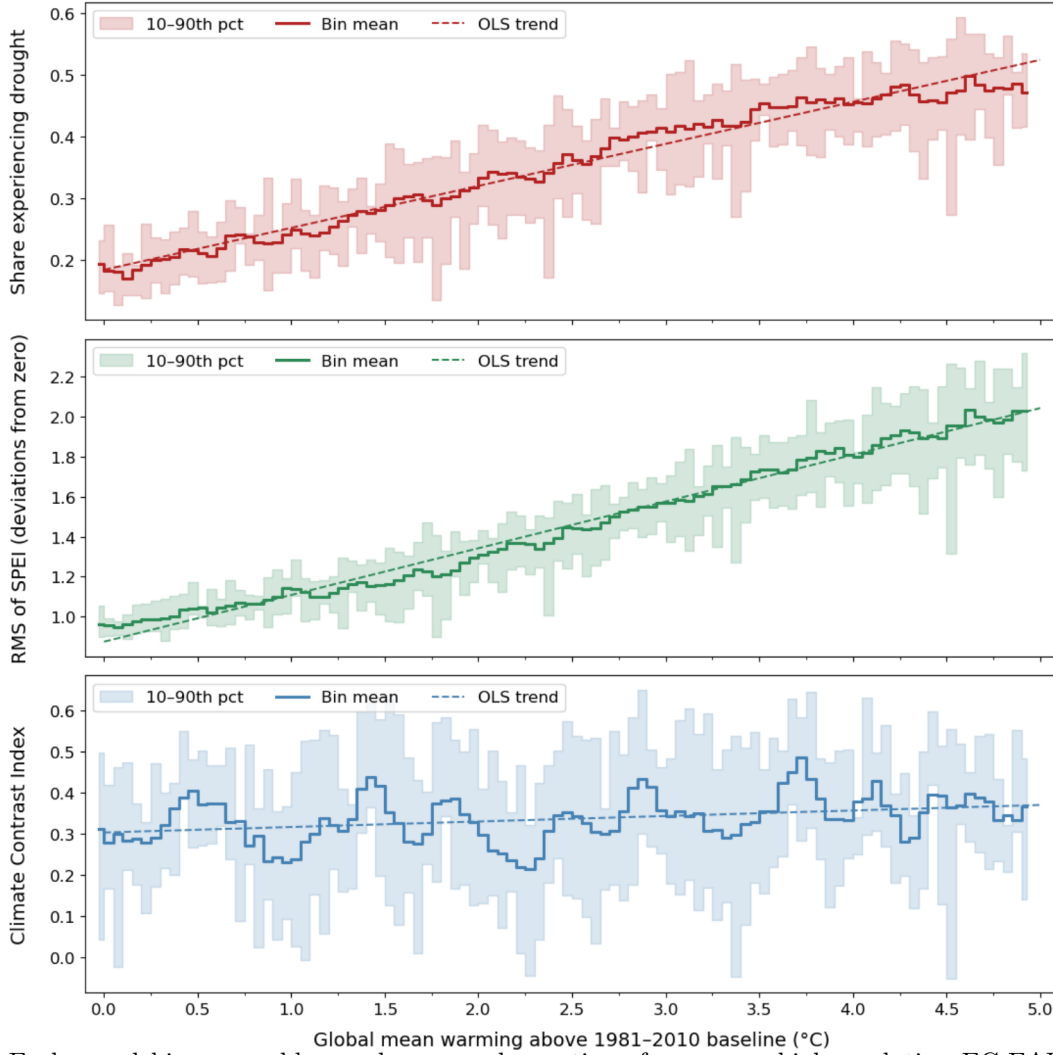
7.2 Climate Change and the Risk-Signal Gap

A final implication of the paper is that climate change may widen the gap between the risk households face and the signal they respond to. If warming raised climate contrast alongside climate variability, households’ experience-based responses could naturally scale with rising risk. The projections show that this scaling does not occur.

I use seven high-resolution EC-EARTH3-HR projections under RCP8.5, spanning 1975–2100 and driven by boundary conditions from seven CMIP5 models (Naumann et al., 2018; Naumann, Alfieri and Wyser, 2026). For each ensemble member and year, I construct drought incidence, the root-mean-square deviation of SPEI from zero, and five-year rolling climate contrast, and relate these measures to an interpolated global-warming index relative to the 1981–2010 baseline (Appendix F details the construction). Figure 4 plots the results.

Objective climate risk rises sharply with warming. Between 0°C and 5°C, the share of grid cells experiencing drought roughly triples, while the root-mean-square deviation of SPEI more than doubles. Climate contrast, however, remains essentially flat. Warming increases the amplitude of climate shocks without increasing the relative year-to-year reversals that generate contrast.

Figure 4: Climate Change Raises Objective Risk Measures, but Not Climate Contrast



Notes: Each panel bins ensemble-member-year observations from seven high-resolution EC-EARTH3-HR projections under RCP8.5 (1975–2100) by an interpolated global-warming index relative to the 1981–2010 baseline, in 0.05°C bins. The top panel plots the latitude-weighted share of grid cells with 12-month SPEI below -1 (drought). The middle panel plots the root-mean-square deviation of SPEI from zero within each member-year. The bottom panel plots the latitude-weighted mean of the five-year rolling Climate Contrast Index. Shaded regions show the 10th–90th percentile range within each bin; the dashed line is the OLS trend across the full 0–5°C range. Appendix F details construction.

The implication is a widening risk-signal gap. If households continue to respond primarily to climate contrast, as the historical evidence indicates, precautionary behavior need not keep pace with the rise in objective climate risk. The projections therefore suggest that warming can increase the risks households face without strengthening the signal that currently triggers self-insurance. This extrapolation assumes that households’ reference points and their behavioral response to contrast remain stable as the climate moves beyond the historical range, an assumption the present data cannot test.

8 Conclusion

This paper shows that the climate statistic households respond to is not the one that best predicts the risk they face. Climate variability predicts future drought and volatility but is weakly related to precautionary saving. Climate contrast—the ordering of recent realizations—predicts households’ assessed exposure and precautionary saving, yet carries little information about future climate risk. The recency-weighted framework makes this distinction explicit: climate histories with the same mean and variance can generate different responses depending on the order in which they are experienced.

In Tanzania, climate contrast is associated with greater emergency saving, stronger asset accumulation, and fewer food shortages, with the strongest responses among households most exposed to climate-dependent income and with the thinnest initial buffers. The same saving pattern appears across 29 low- and middle-income countries. Climate variability, by contrast, erodes household assets. These results help explain why conventional measures of climate variability can be weakly or even negatively related to observed saving: greater risk can simultaneously strengthen the motive to self-insure and weaken households’ capacity to do so.

The data do not directly measure beliefs about future risk: the survey records retrospective assessments of climate exposure, and the cross-country evidence serves primarily as a test of external validity. Nor can the climate projections establish that households’ reference points will remain unchanged under unprecedented warming. But the central result does not depend on these extrapolations: households systematically respond to a feature of recent experience that contains little information about future risk. This distinction becomes increasingly consequential as warming raises objective climate risk while leaving climate contrast largely unchanged. Effective adaptation therefore requires both better signals of changing risk and financial buffers that allow households to act on them.

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Online Appendix for “When the Right Signal Fails: Climate Risk, Precautionary Saving, and the Limits of Self-Insurance”

A Theoretical Derivations

This appendix provides the derivations underlying Proposition 1, Equation (5), and Corollary 1. Section A.1 shows that mean-based and sequential surprise are limiting cases of the recency-weighted benchmark introduced in Definition 1. Section A.2 then derives the decomposition of sequential surprise into dispersion and ordering components.

A.1 Recency-Weighted Benchmarks and Limiting Cases

Recall from Definition 1 that

$$B_t(\delta) = (1 - \delta) \sum_{k=1}^{\infty} \delta^{k-1} W_{t-k}, \quad \delta \in [0, 1), \quad (10)$$

with

$$S_t(\delta) = |W_t - B_t(\delta)|. \quad (11)$$

Suppose $\{W_t\}$ is covariance stationary and mean-square ergodic, with Gaussian marginal distribution $N(\mu_W, \sigma_W^2)$. I first show that

$$\lim_{\delta \rightarrow 0} B_t(\delta) = W_{t-1}, \quad (12)$$

$$\lim_{\delta \rightarrow 1^-} B_t(\delta) = \mu_W \quad (13)$$

in mean square.

For the short-memory limit,

$$B_t(\delta) = (1 - \delta) (W_{t-1} + \delta W_{t-2} + \delta^2 W_{t-3} + \cdots). \quad (14)$$

As $\delta \rightarrow 0$, the weight on W_{t-1} converges to one and all remaining weights vanish. Hence

$$\lim_{\delta \rightarrow 0} B_t(\delta) = W_{t-1}. \quad (15)$$

It follows that

$$\lim_{\delta \rightarrow 0} S_t(\delta) = |W_t - W_{t-1}|, \quad (16)$$

and L^2 convergence implies L^1 convergence, so

$$\lim_{\delta \rightarrow 0} E[S_t(\delta)] = E[|W_t - W_{t-1}|]. \quad (17)$$

For the long-memory limit, $B_t(\delta)$ is an Abel, or geometrically weighted, average of the past, with weights that become increasingly diffuse as $\delta \rightarrow 1^-$. Mean-square ergodicity and the Abelian ergodic theorem therefore imply

$$\lim_{\delta \rightarrow 1^-} B_t(\delta) = \mu_W \quad (18)$$

in L^2 . Thus,

$$\lim_{\delta \rightarrow 1^-} S_t(\delta) = |W_t - \mu_W|, \quad (19)$$

and, by the reverse triangle inequality and Cauchy–Schwarz,

$$\lim_{\delta \rightarrow 1^-} E[S_t(\delta)] = E[|W_t - \mu_W|]. \quad (20)$$

Finally, because $W_t - \mu_W \sim N(0, \sigma_W^2)$,

$$E[|W_t - \mu_W|] = \sigma_W \sqrt{\frac{2}{\pi}}. \quad (21)$$

Hence mean-based and sequential surprise are the long- and short-memory limits, respectively, of the same recency-weighted benchmark, establishing Proposition 1. $\square$

A.2 Sequential Surprise Decomposition

I next derive Equation (5). Under the Gaussian benchmark above, let

$$\Delta_t \equiv W_t - W_{t-1}.$$

Since (W_t, W_{t-1}) is jointly Gaussian, Δ_t is normally distributed with mean zero and variance

$$\begin{aligned} \text{Var}(\Delta_t) &= \text{Var}(W_t) + \text{Var}(W_{t-1}) - 2\text{Cov}(W_t, W_{t-1}) \\ &= 2\sigma_W^2(1 - \rho), \end{aligned} \quad (22)$$

where stationarity implies $\text{Cov}(W_t, W_{t-1}) = \rho\sigma_W^2$.

For a mean-zero normal random variable $X \sim N(0, \sigma_X^2)$, $E[|X|] = \sigma_X \sqrt{2/\pi}$. Applying

this identity to Δ_t gives

$$E[|W_t - W_{t-1}|] = \sqrt{2\sigma_W^2(1-\rho)}\sqrt{\frac{2}{\pi}} \quad (24)$$

$$= \sigma_W \sqrt{\frac{2}{\pi}} \times \sqrt{2(1-\rho)}, \quad (25)$$

which is Equation (5).

The first factor, $\sigma_W \sqrt{2/\pi}$, is identical to mean-based surprise and captures dispersion. The second, $\sqrt{2(1-\rho)}$, captures temporal ordering: it increases as consecutive realizations become less persistent and more strongly reversing. Dividing by σ_W gives

$$\frac{E[|W_t - W_{t-1}|]}{\sigma_W} = 2\sqrt{\frac{1-\rho}{\pi}}, \quad (26)$$

which establishes the Climate Contrast Index in Corollary 1. $\square$

B Data and Descriptive Statistics

Table B.1: Household Responses to Negative Income Shocks (Tanzania NPS)

Coping strategy	Share of households (%)
Own savings	46
Transfers from relatives or friends	21
Reduced food consumption	6
Increased labor supply	4
Sold livestock	3
Obtained credit	2
Government or NGO assistance	2
Sold assets	1
Migration or sending children elsewhere	1
Other	14

Notes: The table reports the share of households reporting the primary coping strategy among those that took any action to cope with negative income shocks over the past five years. “Other” includes spiritual or religious responses (e.g. prayer; approximately 7% of households, and unspecified responses; 6%). Data come from four rounds of the harmonized Tanzania National Panel Survey 2008-2014. Response categories are rounded to the nearest whole number.

Table B.2: Summary statistics for the full and rural samples in the FinScope data

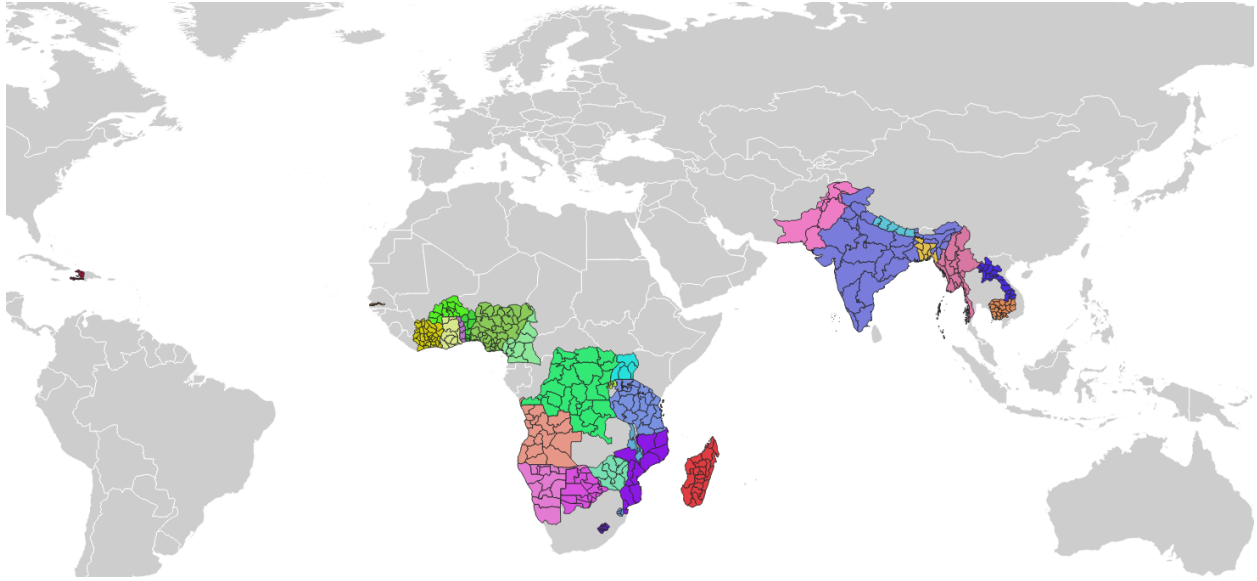
	Full sample		Rural sample	
	Mean	SD	Mean	SD
Saved in past 12 months	0.613	0.49	0.605	0.49
Have loans	0.314	0.46	0.333	0.47
Have insurance	0.149	0.36	0.133	0.34
SPEI (12m)	-0.190	0.91	-0.080	0.89
Climate Variability (5y)	0.024	0.98	-0.006	0.95
Climate Contrast (5y)	-0.017	0.97	-0.021	0.96
Age category	3.136	1.41	3.145	1.41
Education level	2.327	1.07	2.156	1.00
Urban	0.347	0.48	0.000	0.00
Female	0.546	0.50	0.537	0.50
Year of observation	2014.893	3.66	2014.724	3.48
Observations	220,986		144,191	

Note: Ages are reported as quintile groups, ranging from 16-24 years (1), 25-34 years (2), 35-44 years (3), 45-60 years (4), and 61+ years (5). Education is categorized as: no education or incomplete primary school (1), completed primary school (2), completed secondary school (3), vocational training (4), and completed tertiary education (5).

Table B.3: List of countries and waves in the FinScope data

Country	Years
Angola	2022
Bangladesh	2016
Benin	2018
Botswana	2020
Burkina Faso	2016
Cambodia	2015
Cameroon	2017
Cote d'Ivoire	2016
Democratic Republic of Congo	2014
Eswatini	2014, 2018
Gambia	2019
Ghana	2010, 2021
Haiti	2018
India	2015
Laos	2014
Lesotho	2021
Madagascar	2016
Malawi	2014
Mozambique	2014, 2015, 2019
Myanmar	2018
Namibia	2011, 2017
Nepal	2015
Nigeria	2016
Pakistan	2008
Rwanda	2008, 2012, 2016
Tanzania	2006, 2009, 2013, 2017
Togo	2016
Uganda	2006, 2009, 2013, 2018
Zimbabwe	2014, 2022

Figure B.1: Geographic Coverage of the FinScope National Survey Data



Notes: Each country in the dataset is represented by a unique color. The region (admin 1) boundaries within each country indicate the aggregation level of the FinScope household finance data.

C Climate Shocks and Agricultural Production

I use global data on annual agricultural production and yields from FAOSTAT ([Food and Agriculture Organization of the United Nations, 2024](#)) to validate the interpretation of climate conditions as shifters of agricultural income (Section 3.5). The analysis focuses on six staple food crops that account for a large share of caloric intake in low-income countries: maize, rice, cassava, sorghum, millet, and wheat. For rice, I restrict attention to rainfed production. The sample covers the period 2000–2022.

FAOSTAT reports production at the country level. To map sub-national climate variation to country-level outcomes, I combine these data with the Spatial Production Allocation Model (SPAM) ([You et al., 2014](#)), which provides gridded estimates of crop-specific production patterns. These spatial distributions are treated as predetermined and are used to aggregate climate exposure to the country–crop–year level using crop-area weights, so that local climate realizations affect national production in proportion to where crops are grown.

The main outcome variables are *log crop production* and *log yield* (production per harvested area). Climate exposure is measured using contemporaneous 12-month SPEI values, which capture cumulative water availability over the relevant agricultural year. Table C.1 reports the results discussed in Section 3.5.

Table C.1: Climate uncertainty and agricultural production

	Global Sample				FinScope Regions			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Log Production								
SPEI (12m)	0.050*** (0.008)		0.026*** (0.009)	0.029*** (0.009)	0.069*** (0.014)		0.048*** (0.017)	0.054*** (0.016)
Drought (12m)		-0.119*** (0.017)	-0.080*** (0.021)	-0.071*** (0.020)		-0.138*** (0.029)	-0.070* (0.036)	-0.044 (0.033)
Climate Variability (5y)				-0.016 (0.012)				-0.040 (0.026)
Climate Contrast (5y)				0.001 (0.006)				0.005 (0.013)
Panel B: Log Yield								
SPEI (12m)	0.023*** (0.006)		0.014* (0.007)	0.016** (0.007)	0.008 (0.013)		0.009 (0.016)	0.012 (0.016)
Drought (12m)		-0.052*** (0.013)	-0.032* (0.016)	-0.025 (0.016)		-0.009 (0.027)	0.004 (0.032)	0.016 (0.030)
Climate Variability (5y)				-0.009 (0.010)				-0.016 (0.023)
Climate Contrast (5y)				0.005 (0.005)				0.003 (0.014)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Climate controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,181	14,181	14,181	14,181	6,437	6,437	6,437	6,437
Clusters	181	181	181	181	66	66	66	66

Note: *Log Production* is the natural logarithm of crop production (tonnes), and *Log Yield* is the natural logarithm of production per harvested area (tonnes per hectare), measured at the crop–country–year level. Crops included in the analysis are rice, wheat, maize, cassava, sorghum, and millet. *FinScope Regions* indicates countries located in the regions covered by the FinScope survey dataset (Sub-Saharan Africa, South Asia, and Southeast Asia). *SPEI (12m)* is the average Standardized Precipitation–Evapotranspiration Index over the preceding 12 months, aggregated at the country level. *Drought (12m)* is an indicator variable equal to one if SPEI (12m) is below -1 . Climate controls include annual SPEI values for the preceding five years, excluding the most recent 12 months. *Climate Variability (5y)* is defined as the standard deviation of annual SPEI values over the preceding five years. *Climate Contrast (5y)* is defined as the five-year rolling mean of absolute year-to-year changes in SPEI, normalized by (divided by) Climate Variability (5y). Regressions are weighted by each crop’s contribution to total agricultural production within a country. Standard errors are clustered at the country level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.2: Sensitivity to measurement error in the SPEI on crop production, for countries in the FinScope regions

	Dependent variable: Log Production			
	Rainfall station density		Temperature station density	
	Above median	Below median	Above median	Below median
SPEI (12m)	0.060*** (0.020)	0.045 (0.029)	0.074*** (0.022)	0.033 (0.026)
Drought (12m)	-0.024 (0.045)	-0.075 (0.048)	-0.005 (0.046)	-0.090* (0.046)
SPEI Variability (5y)	-0.093 (0.095)	-0.147 (0.112)	-0.108 (0.090)	-0.172 (0.135)
Climate Contrast (5y)	0.064 (0.075)	-0.094 (0.124)	0.067 (0.087)	-0.062 (0.093)
Mean stations	5.64	1.87	7.84	6.55
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Climate controls	Yes	Yes	Yes	Yes
Observations	3,128	3,309	3,152	3,285
Clusters	33	33	33	33

Note: *Log Production* is the log of production (tonnes) for each crop, country, and year. Crops included in the analysis are: rice, wheat, maize, cassava, sorghum and millet. Only countries located in the regions of the FinScope survey dataset are included in this analysis (Sub-Saharan Africa, South Asia and Southeast Asia). *Rainfall* and *Temperature station density* are the average number of stations used in the interpolation of SPEI values for each country over the time period. *SPEI (12m)* is the average Standardized Precipitation–Evapotranspiration Index over the preceding 12 months, aggregated at the country level. *Drought (12m)* is an indicator variable equal to one if SPEI (12m) is below -1 . Climate controls include annual SPEI values for the preceding five years, excluding the most recent 12 months. *Climate Variability (5y)* is defined as the standard deviation of annual SPEI values over the preceding five years. *Climate Contrast (5y)* is defined as the five-year rolling mean of absolute year-to-year changes in SPEI, normalized by (divided by) Climate Variability (5y). Regressions are weighted by each crop’s contribution to total agricultural production within a country. Standard errors are clustered at the country level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

D Additional Results and Heterogeneity

Table D.1: Climate Contrast Does Not Predict Migration

	Net migration rate	Any move	Moved far	Moved locally
	(1)	(2)	(3)	(4)
SPEI (12m)	-0.241** (0.105)	0.008 (0.019)	0.013 (0.020)	-0.005 (0.018)
Climate Variability (5y)	0.063 (0.256)	0.014 (0.021)	0.014 (0.019)	0.000 (0.017)
Climate Contrast (5y)	0.112 (0.165)	-0.002 (0.009)	0.001 (0.009)	-0.003 (0.009)
Sample	FinScope regions	Tanzania LSMS	Tanzania LSMS	Tanzania LSMS
Type of FE	Region (Admin 1)	Household	Household	Household
Year FE	Yes	Yes	Yes	Yes
Observations	6,363	12,712	12,712	12,712
Clusters	458	139	139	139

Note: *Net migration rate* is the number of net migrants per 1000 inhabitants in each region and year, where positive value indicates net in-migration and negative values net out-migration. *Ever moved* is a binary variable equal to 1 if the household relocated between survey waves. *SPEI (12m)* is the average SPEI value over the past 12 months aggregated at the region (admin 1) level for column (1) and at the district (admin 2) level for columns (2)-(4). Climate controls include yearly SPEI values for the last 5 years, excluding the last 12 months. *Climate Variability (5y)* is the standard deviation of last 5 years' SPEI values. *Climate Contrast (5y)* is the average absolute difference in SPEI over the last 5 years, normalized by (divided by) Climate Variability (5y). Standard errors are clustered at the region (admin 1) level for column (1) and at the district (admin 2) level for columns (2)-(4). * p<0.1, ** p<0.05, *** p<0.01.

Table D.2: Climate uncertainty and saving behavior, by labor skill level

	Saved in past 12 months			
	Unskilled (1)	Low skill (2)	Medium skill (3)	High skill (4)
SPEI (12m)	0.010 (0.018)	0.010 (0.015)	−0.000 (0.012)	−0.003 (0.017)
Climate Variability (5y)	−0.018 (0.014)	−0.007 (0.011)	−0.008 (0.008)	0.022 (0.014)
Climate Contrast (5y)	0.036*** (0.012)	0.027*** (0.007)	0.009 (0.006)	0.010 (0.010)
Sample	Rural	Rural	Rural	Rural
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
HH controls	Yes	Yes	Yes	Yes
Observations	41,403	56,181	42,466	4,140
Clusters	360	365	366	283

Note: *Unskilled labor* is defined as having not completed primary education, *low skill* as having completed only primary education, *medium skill* as having completed secondary education and/or received vocational training, and *high skill* as having completed tertiary education. *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the region (admin 1) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five and thirty years, as well as Climate Variability and climate contrast measured over the previous thirty years. Household controls include age category, an urban indicator, and a female indicator. Standard errors are clustered at the region (admin 1) level. * p<0.1, ** p<0.05, *** p<0.01.

Table D.3: Climate Contrast and Saving, by Gender

	Saved in past 12 months	
	Men (1)	Women (2)
SPEI (12m)	−0.002 (0.013)	0.001 (0.012)
SPEI Variability (5y)	−0.014 (0.009)	−0.017* (0.009)
Climate Contrast (5y)	0.026*** (0.008)	0.027*** (0.008)
Sample	Rural	Rural
Country FE	Yes	Yes
Year FE	Yes	Yes
HH controls	Yes	Yes
Observations	67,425	78,077
Clusters	373	373

Note: *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the region (admin 1) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five and thirty years, as well as Climate Variability and climate contrast measured over the previous thirty years. Household controls include age category, education level, and an urban indicator. Standard errors are clustered at the region (admin 1) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.4: Climate Contrast and Stated Importance of Mobile Money Services Used

	Most important out of used services							
	Buy airtime for yourself	Buy airtime for someone else	Send money	Receive money	Paid for a good/service	Save for an emergency	Save for everyday expenses	Save for a large purchase
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: District Fixed Effects								
SPEI (12m)	0.026** (0.012)	0.009*** (0.003)	0.008 (0.033)	-0.031 (0.032)	0.015 (0.017)	-0.009 (0.016)	-0.020 (0.018)	0.001 (0.003)
SPEI Variability (5y)	-0.014 (0.011)	0.001 (0.003)	-0.006 (0.022)	0.048** (0.022)	-0.008 (0.010)	-0.012 (0.009)	-0.006 (0.010)	-0.003 (0.002)
Climate Contrast (5y)	0.006 (0.007)	-0.000 (0.002)	-0.036** (0.016)	-0.000 (0.014)	0.003 (0.004)	0.015** (0.007)	0.012** (0.006)	-0.000 (0.001)
Observations	7,325	7,325	7,325	7,325	7,325	7,325	7,325	7,325
Panel B: Household Fixed Effects								
SPEI (12m)	0.018 (0.029)	0.005 (0.005)	0.026 (0.052)	-0.034 (0.048)	0.032 (0.024)	-0.022 (0.021)	-0.022 (0.022)	-0.002 (0.008)
SPEI Variability (5y)	-0.021 (0.035)	0.004 (0.007)	0.021 (0.048)	0.029 (0.046)	-0.023 (0.016)	-0.007 (0.016)	-0.007 (0.022)	0.004 (0.007)
Climate Contrast (5y)	0.025 (0.016)	-0.001 (0.004)	-0.056* (0.031)	0.001 (0.029)	0.002 (0.007)	0.023 (0.015)	0.009 (0.012)	-0.003 (0.004)
Observations	2,692	2,692	2,692	2,692	2,692	2,692	2,692	2,692

Note: *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five years. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.5: Climate Contrast and Basic Usage of Mobile Money

	Basic usage of mobile money services					
	Use mobile money		Received money		Sent money	
	(1)	(2)	(3)	(4)	(5)	(6)
SPEI (12m)	0.017 (0.023)	0.007 (0.024)	−0.054** (0.024)	−0.058 (0.046)	−0.015 (0.039)	0.006 (0.048)
SPEI Variability (5y)	−0.002 (0.016)	0.000 (0.021)	−0.006 (0.020)	−0.021 (0.035)	0.016 (0.024)	0.042 (0.039)
Climate Contrast (5y)	0.017 (0.013)	0.016 (0.016)	0.030*** (0.011)	0.032* (0.018)	−0.015 (0.013)	−0.006 (0.020)
FE specification	District FE	Household FE	District FE	Household FE	District FE	Household FE
Survey Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	26,821	21,343	7,353	2,714	7,353	2,714
Clusters	140	140	135	119	135	119

Note: *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five years. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.6: Climate Contrast and Consumption Adjustment Margins

	Log non-food expenditures			
	Baseline		State-contingent	
	(1)	(2)	(3)	(4)
SPEI (12m)	0.120 (0.146)	0.057 (0.165)	-0.114 (0.123)	-0.145 (0.157)
Climate Variability (5y)	0.142 (0.121)	0.260 (0.160)	0.382** (0.149)	0.481** (0.198)
Climate Variability (5y) $\times$ SPEI _{12m}			0.275 (0.173)	0.253 (0.204)
Climate Contrast (5y)	0.153** (0.060)	0.258*** (0.082)	-0.284*** (0.104)	-0.223* (0.132)
Climate Contrast (5y) $\times$ SPEI _{12m}			-0.465*** (0.113)	-0.477*** (0.151)
Type of FE	District FE	Household FE	District FE	Household FE
Survey Year FE	Yes	Yes	Yes	Yes
Observations	26,821	21,343	26,821	21,343
Clusters	140	140	140	140

Note: *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five years. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.7: Climate Contrast and Perceived Severe Climate Shocks, by Rural and Urban Households

	Reports severe climate shock			
	(1)	(2)	(3)	(4)
SPEI (12m)	0.052* (0.027)	0.049 (0.042)	0.065** (0.028)	0.045 (0.050)
Climate Variability (5y)	0.049** (0.021)	0.049 (0.031)	-0.015 (0.020)	-0.022 (0.032)
Climate Contrast (5y)	0.020** (0.009)	0.024* (0.014)	0.016 (0.010)	0.004 (0.018)
Sample	Rural	Rural	Urban	Urban
Type of FE	District FE	Household FE	District FE	Household FE
Survey Year FE	Yes	Yes	Yes	Yes
Observations	14,491	10,553	8,180	5,050
Clusters	136	130	111	105

Note: *Reports climate shock severity* is an indicator equal to one if the household reports experiencing a severe climate shock (drought or flood) during the previous five years, and zero otherwise. *Abs(SPEI 12m)* is the absolute value of the Standardized Precipitation–Evapotranspiration Index over the past 12 months, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include the absolute values of annual SPEI realizations for each of the previous five years, as well as survey year and month fixed effects. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.8: Climate uncertainty and saving behavior, by current climate conditions and type of saving

	Saved in past 12 months, by type					
	Wet year			Dry year		
	Formal	Informal	Family and friends	Formal	Informal	Family and friends
SPEI (12m)	0.004 (0.006)	0.045** (0.020)	0.019 (0.022)	-0.003 (0.009)	0.003 (0.021)	0.008 (0.025)
Climate Variability (5y)	-0.001 (0.004)	-0.010 (0.018)	0.015 (0.016)	0.005 (0.007)	-0.034** (0.014)	0.008 (0.018)
Climate Contrast (5y)	0.004 (0.003)	0.032** (0.014)	0.029** (0.012)	0.010** (0.005)	-0.017* (0.010)	0.005 (0.012)
Sample	Rural	Rural	Rural	Rural	Rural	Rural
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
HH controls	Yes	Yes	Yes	Yes	Yes	Yes
Mean DV	0.111	0.320	0.325	0.133	0.244	0.311
Observations	66,050	66,050	66,050	78,141	78,141	78,141
Clusters	164	164	164	237	237	237

Note: *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the region (admin 1) level. *Wet year* refers to observations with $SPEI(12m) > 0$, while *Dry year* refers to observations with $SPEI(12m) \leq 0$. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five and thirty years, as well as Climate Variability and climate contrast measured over the previous thirty years. Household controls include age category, education level, an urban indicator, and a female indicator. Standard errors are clustered at the region (admin 1) level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

E Additional Robustness Checks

Table E.1: Climate Contrast and Climate Shock Perception Using an Extensive Set of Climate Controls

	Reports severe climate shock							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
SPEI (12m)	0.059*** (0.021)	0.057*** (0.021)	0.046** (0.022)	0.042* (0.022)	0.043 (0.034)	0.038 (0.033)	0.019 (0.038)	0.012 (0.038)
Climate Variability (5y)	0.036** (0.017)	0.037* (0.022)	0.036** (0.017)	0.042** (0.021)	0.034 (0.024)	0.029 (0.031)	0.039 (0.025)	0.039 (0.031)
Climate Contrast (5y)	0.023*** (0.007)	0.023*** (0.007)	0.022*** (0.007)	0.021*** (0.007)	0.024** (0.011)	0.024** (0.011)	0.022** (0.011)	0.022* (0.011)
Type of FE	District	District	District	District	HH	HH	HH	HH
Baseline climate controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
5-year max/min SPEI	No	Yes	No	Yes	No	Yes	No	Yes
Drought/flood counts	No	No	Yes	Yes	No	No	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,680	22,680	22,680	22,680	17,052	17,052	17,052	17,052
Clusters	140	140	140	140	140	140	140	140

Note: *Reports climate shock severity* is an indicator equal to one if the household reports experiencing a severe climate shock (drought or flood) during the previous five years, and zero otherwise. *Abs(SPEI 12m)* is the absolute value of the Standardized Precipitation–Evapotranspiration Index over the past 12 months, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include the absolute values of annual SPEI realizations for each of the previous five years, as well as survey year and month fixed effects. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table E.2: Climate Contrast and Saving Behavior Using Alternative Exposure Windows

	(1) Baseline 5y	(2) EW $h = 1.5$	(3) EW $h = 2$	(4) EW $h = 3$	(5) EW $h = 4$
Panel A: Tanzania FinScope					
Climate Variability	0.027 (0.020)	−0.003 (0.023)	0.001 (0.023)	0.008 (0.024)	0.011 (0.025)
Climate Contrast	0.029** (0.012)	0.037** (0.017)	0.039** (0.016)	0.039*** (0.014)	0.038*** (0.014)
Mean DV	0.296	0.296	0.296	0.296	0.296
Observations	7,205	7,205	7,205	7,205	7,205
Clusters	129	129	129	129	129
Panel B: Tanzania LSMS					
Climate Variability	0.041 (0.036)	0.026 (0.037)	0.014 (0.035)	−0.000 (0.032)	−0.007 (0.030)
Climate Contrast	0.025 (0.024)	0.044** (0.022)	0.055** (0.022)	0.066*** (0.023)	0.072*** (0.024)
Memory weighting	Equal	Exp.	Exp.	Exp.	Exp.
Half-life (years)	—	1.5	2	3	4
Mean DV	0.285	0.285	0.285	0.285	0.285
Observations	2,714	2,714	2,714	2,714	2,714
Clusters	119	119	119	119	119

Note: *Climate Variability* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). The exponentially weighted measures assign declining weights to older annual SPEI realizations, with the indicated half-life, and apply the same weighting scheme to the year-to-year changes entering Climate Contrast and to the SPEI realizations used to construct the corresponding Climate Variability measure. The exponentially weighted measures use annual SPEI realizations over a 10-year window. All regressions control for current SPEI and average SPEI over the previous five years. The Tanzania FinScope regressions additionally include age, education, household-characteristic, urban-residence, and gender controls. FinScope regressions include district and survey-wave fixed effects, while NPS regressions include household and survey-round fixed effects. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table E.3: Climate Contrast and Saving Behavior Using an Extensive Set of Climate Controls

	Saved in past 12 months			
	(1)	(2)	(3)	(4)
SPEI (12m)	0.005 (0.012)	0.005 (0.011)	0.004 (0.011)	0.000 (0.013)
Climate Variability (5y)	-0.010 (0.009)	-0.005 (0.012)	-0.009 (0.009)	-0.003 (0.012)
Climate Contrast (5y)	0.024*** (0.007)	0.025*** (0.007)	0.024*** (0.007)	0.025*** (0.007)
Sample	Rural	Rural	Rural	Rural
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
HH controls	Yes	Yes	Yes	Yes
Baseline climate controls	Yes	Yes	Yes	Yes
Drought count (SPEI < -1) _{5y}	No	Yes	No	Yes
Yearly SPEI lags _{5y}	No	No	Yes	Yes
Observations	144,191	144,191	144,191	144,191
Clusters	368	368	368	368

Note: *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the region (admin 1) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five and thirty years, as well as Climate Variability and climate contrast measured over the previous thirty years. Household controls include age category, education level, an urban indicator, and a female indicator. Standard errors are clustered at the region (admin 1) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table E.4: Tanzania Results Are Robust to Spatially Correlated Climate Shocks

	(1) Baseline	(2) 50km	(3) 100km	(4) 200km	(5) 400km
Panel A: Reports Severe Climate Shock – District FE					
Climate Variability (5y)	0.036** (0.017)	0.036** (0.017)	0.036* (0.018)	0.036** (0.018)	0.036*** (0.014)
Climate Contrast (5y)	0.023*** (0.007)	0.023*** (0.006)	0.023*** (0.006)	0.023*** (0.006)	0.023*** (0.005)
Observations	22,680	22,680	22,680	22,680	22,680
Panel B: Reports Severe Climate Shock – Household FE					
Climate Variability (5y)	0.034 (0.024)	0.034* (0.019)	0.034* (0.020)	0.034* (0.020)	0.034** (0.014)
Climate Contrast (5y)	0.023** (0.011)	0.023*** (0.008)	0.023*** (0.008)	0.023*** (0.008)	0.023*** (0.007)
Observations	17,052	17,052	17,052	17,052	17,052
Panel C: Emergency Saving – District FE					
Climate Variability (5y)	–0.008 (0.019)	–0.008 (0.019)	–0.008 (0.018)	–0.008 (0.018)	–0.008 (0.017)
Climate Contrast (5y)	0.022** (0.011)	0.022** (0.011)	0.022** (0.011)	0.022** (0.011)	0.022** (0.010)
Observations	7,353	7,354	7,354	7,354	7,354
Panel D: Emergency Saving – Household FE					
Climate Variability (5y)	0.035 (0.030)	0.035* (0.021)	0.035* (0.020)	0.035* (0.019)	0.035** (0.017)
Climate Contrast (5y)	0.023 (0.022)	0.023 (0.016)	0.023 (0.017)	0.023 (0.017)	0.023* (0.013)
Observations	2,714	2,714	2,714	2,714	2,714

Note: Panels A–B replicate the district- and household-FE specifications of [Table 1](#) (reported severe climate shock); Panels C–D replicate those of [Table 2](#) (emergency saving). Column (1) reproduces the paper’s baseline specification, with standard errors clustered at the district (admin 2) level. Columns (2)–(5) hold the point estimate fixed at its column-(1) value (numerically identical across columns to several decimal places, since the underlying point estimate does not depend on the standard-error correction) and instead report Conley ([Conley, 1999](#)) spatial HAC standard errors, using a Bartlett (linearly decaying) kernel, at distance cutoffs of 50, 100, 200, and 400 km. Household coordinates are averaged to the district level, since climate exposure and the district/household fixed effects used throughout are themselves assigned at that level; households in districts without available coordinates are excluded (roughly 3% of the estimation sample, so observation counts differ marginally from the main-text tables). Estimation uses the `acreg` Stata command. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table E.5: Sensitivity to measurement error in the SPEI on saving behavior

	Dependent variable: Saved in past 12 months			
	Rainfall station density		Temperature station density	
	Above median	Below median	Above median	Below median
SPEI (12m)	0.004 (0.016)	0.011 (0.016)	0.018 (0.017)	-0.004 (0.015)
SPEI Variability (5y)	-0.020 (0.014)	0.003 (0.009)	-0.017 (0.016)	-0.004 (0.010)
Climate Contrast (5y)	0.026*** (0.009)	0.005 (0.008)	0.028** (0.011)	0.019** (0.008)
Mean no. stations	5.11	2.18	7.86	6.33
Sample	Rural	Rural	Rural	Rural
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
HH controls	Yes	Yes	Yes	Yes
Observations	73,438	72,064	62,109	83,393
Clusters	197	177	212	162

Note: *Rainfall* and *Temperature station density* are the average number of stations used in the interpolation of SPEI values for each country over the time period. *SPEI (12m)* is the 12-month Standardized Precipitation–Evapotranspiration Index, aggregated at the region (admin 1) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include moving averages of SPEI over the previous five and thirty years, as well as Climate Variability and climate contrast measured over the previous thirty years. Household controls include age category, education level, an urban indicator, and a female indicator. Standard errors are clustered at the region (admin 1) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table E.6: Climate Contrast and Non-Climate Shock Perception (Placebo Test)

	Reports severe non-climate shock	
	(1)	(2)
Abs(SPEI 12m)	-0.041** (0.019)	-0.040 (0.025)
SPEI Variability (5y)	0.031** (0.015)	0.038* (0.020)
Climate Contrast (5y)	0.009 (0.007)	0.005 (0.009)
FE specification	District FE	Household FE
Survey Year FE	Yes	Yes
Observations	26,351	20,860
Clusters	140	140

Note: *Reports severe non-climate shock* is an indicator equal to one if the household reports experiencing a severe shock unrelated to climate risk, including: hijacking/robbery/burglary/assault, incarceration, death of a household member, death of another family member, or household breakup, and zero otherwise. *Abs(SPEI 12m)* is the absolute value of the Standardized Precipitation–Evapotranspiration Index over the past 12 months, aggregated at the district (admin 2) level. *Climate Variability (5y)* is the standard deviation of annual SPEI values over the previous five years. *Climate Contrast (5y)* is defined as the average absolute year-to-year change in SPEI over the previous five years, normalized by (divided by) Climate Variability (5y). All regressions include the absolute values of annual SPEI realizations for each of the previous five years, as well as survey year and month fixed effects. Standard errors are clustered at the district (admin 2) level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

F Climate Projections: GCM Methodology

Figure 4 uses the Joint Research Centre’s global monthly SPEI projection archive for 1975–2100 (Naumann, Alfieri and Wyser, 2026), developed by Naumann et al. (2018). The archive contains seven high-resolution projections produced with EC-EARTH3-HR under RCP8.5 and forced by boundary conditions from IPSL-CM5A-LR, GFDL-ESM2M, HadGEM2-ES, EC-EARTH, GISS-E2-H, IPSL-CM5A-MR, and HadCM3LC. The common downscaled grid is approximately 0.35° . I retain land grid cells between 35°S – 35°N and 20°W – 160°E , the broad domain containing the FinScope countries, and use each December observation of 12-month SPEI as the annual value.

For each ensemble member and grid cell, I construct Climate Contrast as the centered five-year rolling mean of absolute annual SPEI changes divided by the centered five-year rolling standard deviation of annual SPEI. Values above three are excluded before aggregation, matching the projection code’s treatment of extreme ratios arising from near-zero denominators. I then form cosine-of-latitude-weighted spatial means. Drought incidence is the corresponding weighted share of grid cells with SPEI below -1 . The middle panel’s variability measure is the root-mean-square deviation of contemporaneous SPEI from zero across grid cells; it is a cross-sectional member-year diagnostic, not the five-year rolling variability regressor used in the household analysis.

The horizontal axis is constructed because the archive contains SPEI but not a global-mean-temperature series. For each forcing model, I interpolate a monotone annual warming path through the published years at which the projection crosses 1.5 , 2 , and 4°C , and anchor the index at zero at the midpoint of the 1981–2010 reference period. I pool the resulting ensemble-member-year observations—approximately 800 across the seven-member ensemble after rolling-window endpoints and warming-range restrictions—into 0.05°C bins. The plotted line is the bin mean, the shaded band is the within-bin 10th–90th percentile range, and the dashed line is an OLS fit to the bin means over 0 – 5°C . Because all members follow RCP8.5 and the warming path is interpolated from crossing years rather than observed directly in these files, the exercise should be read as descriptive evidence across warming levels, not as a scenario-weighted projection.